DSI102 • WEEK 1

Introduction to Digital Technology

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Learning Outcomes

1. **Explain**
   computer platforms and computer networks

2. **Explain**
   digital services available on public networks

3. **Distinguish**
   technology, digital technology, and digital services

4. **Analyze**
   a service through data, software, network, people, and value

Course Description

Computer platforms and networks are the technical foundation for digital services.

- Computer organization: hardware, operating system, and system applications
- Virtualization and cloud computing
- Local area networks, wide area networks, and the Internet
- Principles of network communication and digital services



Foundation → Infrastructure → Services → Society

SECTION 1

Foundational Concepts

What counts as technology, and what changes when it becomes digital.



What Is Technology?

Technology is broader than computers and phones.

Technology is the application of scientific knowledge for practical purposes.

Class prompt

Which examples are technology? Which are digital?

TV	Book	Camera
Pencil	Mobile	Etc.



Technology vs Digital Technology

The difference is not the object; it is how information is represented and processed.

1

Physical tool

Solves a practical task, but information may stay local or analog

2

Digital system

Turns information into data that software can process

3

Digital service

Delivers value through connected platforms and networks

A notebook is technology; a shared cloud document is digital technology; a learning platform is a digital service.

Analog vs Digital

Analog is continuous; digital represents information as discrete data.



Analog

Continuous signal or physical record

sound waves • film • handwriting

Digitization

converts the source into data

scan • encode • sample

Digital

software can store, copy, search, transmit, and analyze

files • apps • dashboards

Data Representation: Why 0 and 1 Matter

Digital systems simplify information into states that machines can process reliably.

1. **Text** — characters encoded as numbers
2. **Image** — pixels with color values
3. **Sound** — samples measured over time
4. **Video** — images + sound + timing

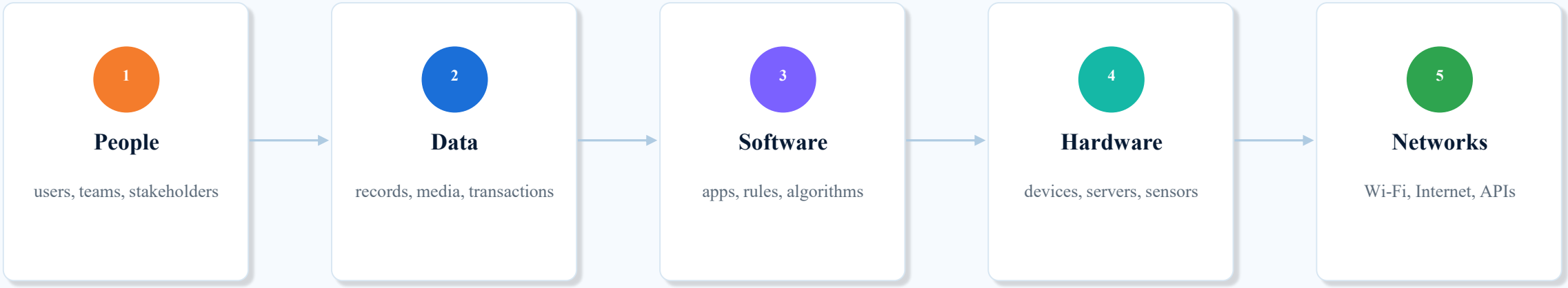
Binary abstraction

0 1 1 0 1 0 0 1
1 0 0 1 0 1 1 0
0 0 1 1 1 0 1 1

Different media can be stored, transmitted, compressed, and protected using the same digital foundation.

Digital Technology = Five Building Blocks

A service is never “just an app.”



Analysis habit: ask what each building block contributes, who manages it, and what can go wrong.

Anatomy of a Digital Service

Most public digital services combine interface, processing, storage, and network access.



- Interface: web, mobile, voice, kiosk
- Logic: rules, algorithms, workflows
- Storage: databases, files, cloud services
- Controls: authentication, authorization, encryption

Example

A banking app is not only a screen; it is a regulated service connected to identity, accounts, transactions, and security systems.

Activity 1: Technology Sorting

Build shared vocabulary before technical detail.

Step 1

Students list 10 tools used today.

Step 2

Classify each as technology, digital technology, or digital service.

Step 3

Discuss borderline cases.



SECTION 2

Types of Digital Technology

Categories are useful, but real-world services often combine multiple types.



A Working Map of Digital Technology

Use categories to ask better questions about systems.

BU

Business

IT, digital marketing, data management, e-commerce

OP

Operational

hardware + software for internal operations

CO

Communication

networks for people and devices

IN

Information

store, send, and retrieve data

ED

Educational

digital tools for teaching and learning

EM

Emerging

AI, blockchain, automation, immersive systems

KEY IDEA

Business Technology

Digital tools help organizations create value, reach customers, and coordinate work.

Examples: information systems, digital marketing, data management, e-commerce.



Operational Technology

Technology used to operate, monitor, and secure organizational processes.

1

Internal systems

intranet, enterprise systems, shared services

2

Control & monitoring

sensors, dashboards, access controls

3

Process reliability

repeatable workflows, logs, alerts

Communication Technology

Digital communication connects users, devices, and services.

- Virtual assistants
- Social media platforms
- Wi-Fi networks
- Bluetooth
- Video conferencing and collaboration tools

Communication technology changes speed, reach, visibility, and coordination.



Information Technology

IT combines hardware, software, and telecommunications to manage information.

1

Store

databases, files, repositories

2

Send

networks, APIs, messaging

3

Retrieve

search, dashboards, reports

Information technology makes data available to the right people and systems at the right time.

Educational Technology

Digital tools can support instruction, interaction, feedback, and learning analytics.



1. **Instruction** — computer-based lessons
2. **Interaction** — quizzes, simulations, discussion
3. **Resources** — online materials and media
4. **Feedback** — analytics, comments, progress tracking

Blockchain Technology

A distributed record can support trust through shared validation and encrypted data.

- Shared ledger
- Cryptographic linking
- Consensus / validation
- Useful for traceability and audit



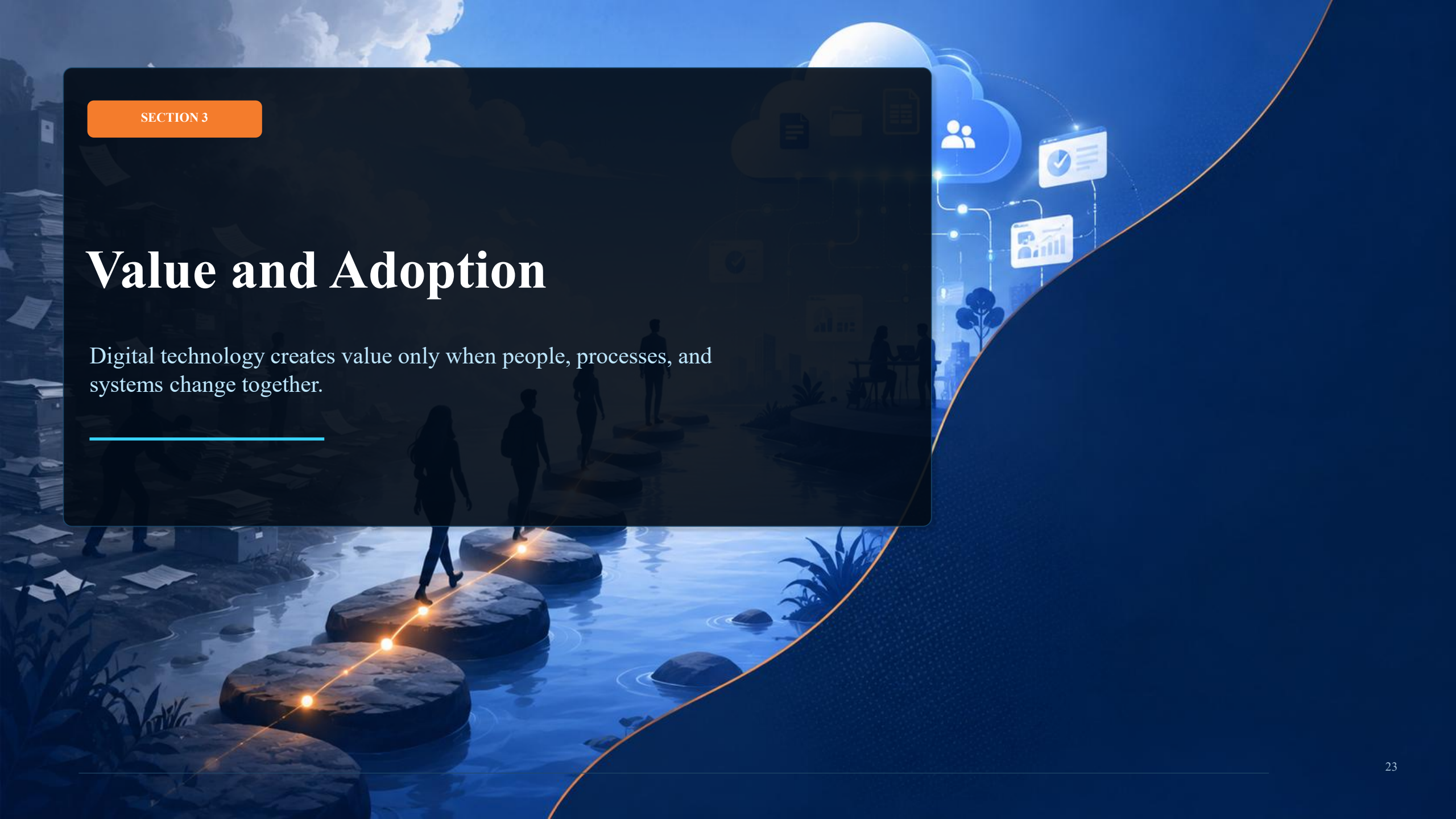
Critical thinking: blockchain is not automatically better; evaluate fit, cost, governance, and privacy.

AI and Automation

AI systems turn data into predictions, recommendations, generated content, or decisions.



- AI can support analysis, search, classification, content generation, and decision support.
- Automation uses rules or models to perform repeatable tasks with less manual effort.
- Human judgment remains important for goals, quality, risk, and ethics.



SECTION 3

Value and Adoption

Digital technology creates value only when people, processes, and systems change together.

Why Digital Technology Matters

It improves what organizations can do, how fast they can do it, and how well they can learn.

1

Streamline

simple, repeatable, automatable work

2

Boost productivity

less friction and better coordination

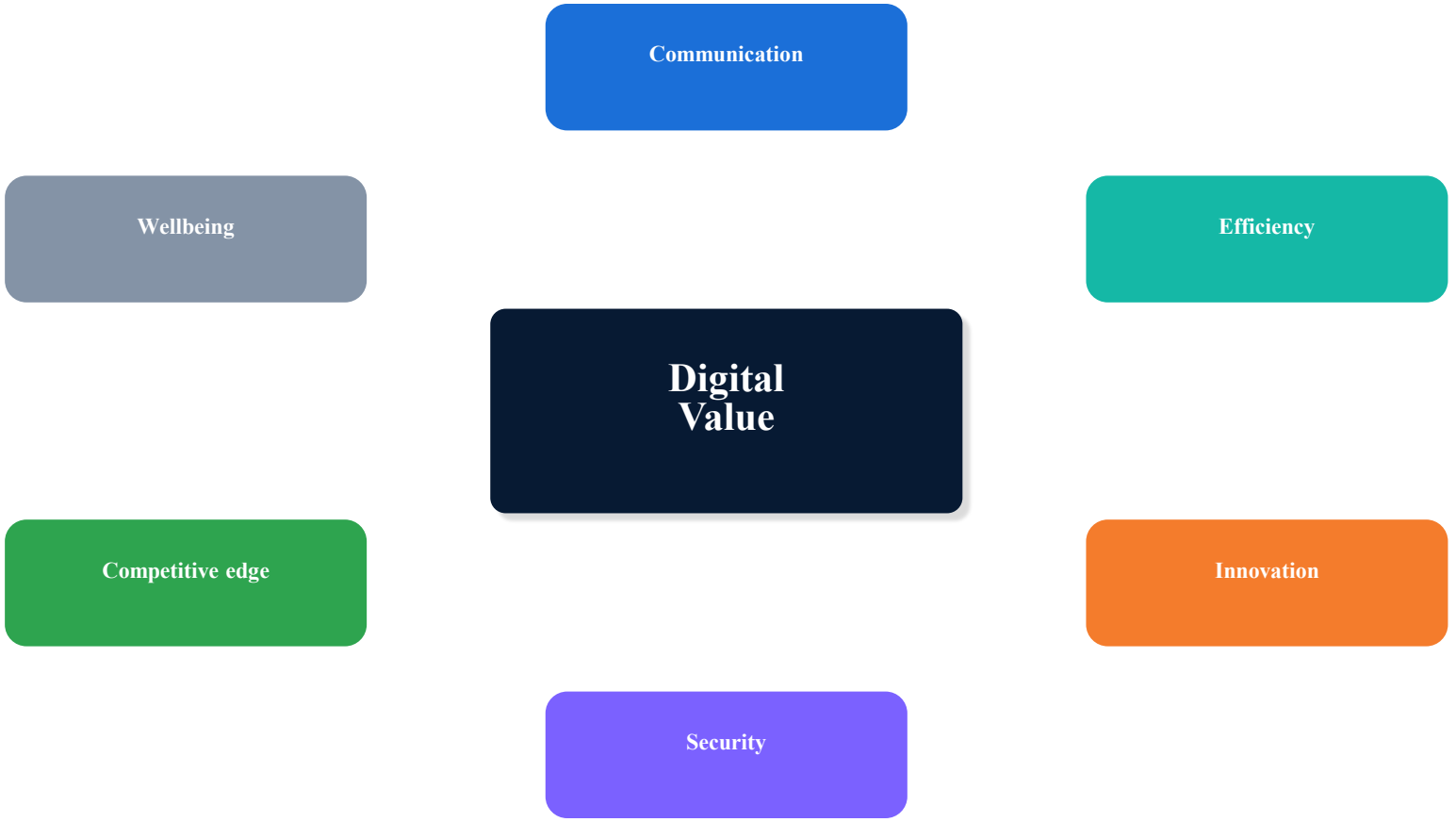
3

Improve experience

faster, more personalized services

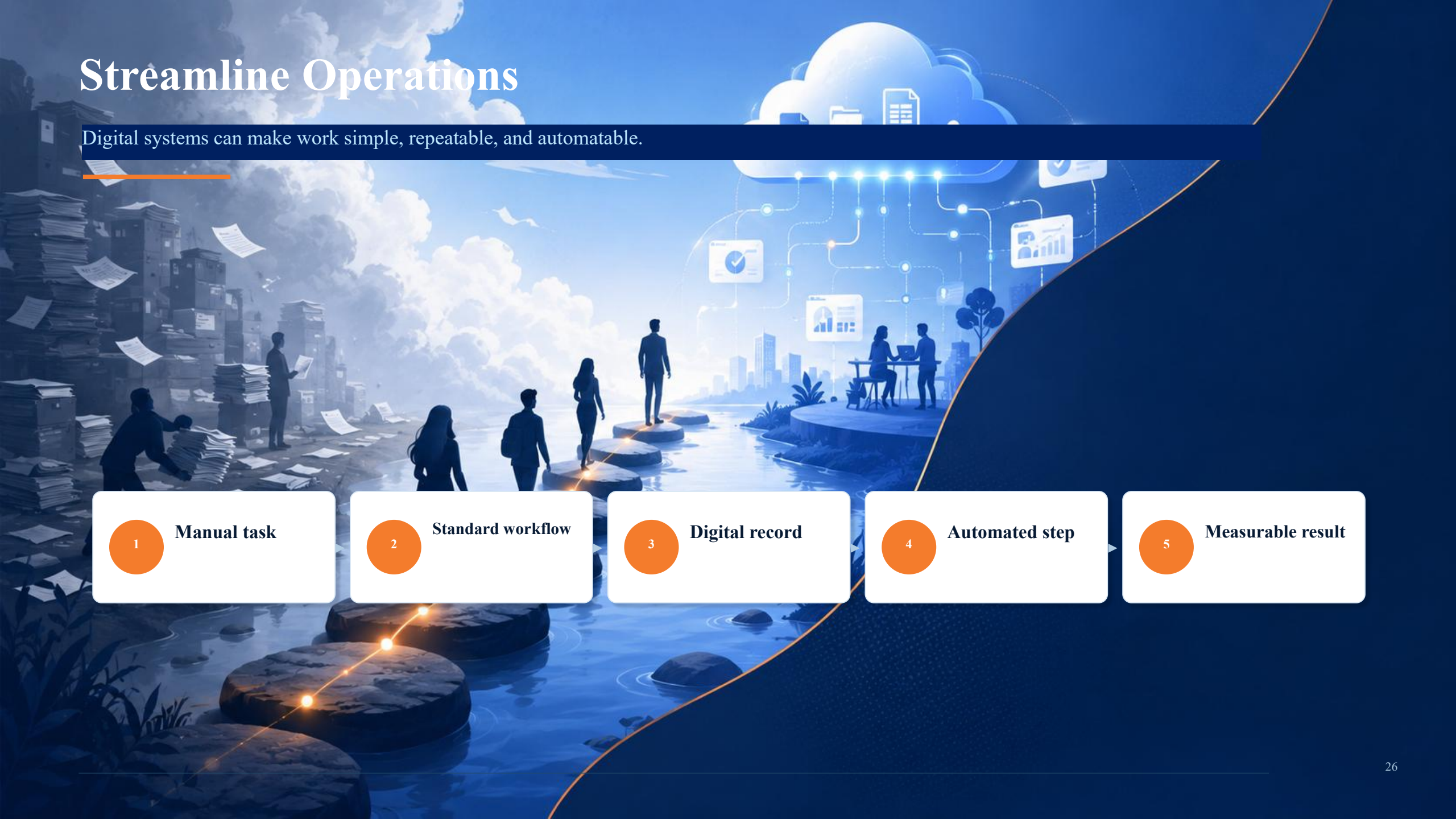
Six Benefits of Using Digital Technology

Benefits are connected: improving one area often affects the others.



Streamline Operations

Digital systems can make work simple, repeatable, and automatable.



1. **Manual task**
2. **Standard workflow**
3. **Digital record**
4. **Automated step**
5. **Measurable result**

Productivity and Efficiency

Productivity gains come from reducing friction, delay, errors, and duplicated effort.

Before

manual entry
searching for files
unclear responsibility

After

shared data
workflow notifications
automated checks

Watch out

bad automation can scale bad process

Customer Experience

Digital services reshape how users discover, access, and receive value.



- Convenience: anytime, anywhere access
- Personalization: recommendations and saved preferences
- Transparency: status, tracking, notifications
- Continuity: service history across channels

Design question

Where does the user feel friction, confusion, delay, or risk?

Data-Driven Decisions

Digital systems leave traces that can be analyzed to improve services.

- Operational data: usage, errors, time, transactions
- Customer data: needs, feedback, behavior patterns
- Decision support: dashboards, alerts, predictions
- Risk: biased, incomplete, or sensitive data



Security and Risk Are Part of Value

Digital services increase reach and efficiency, but they also expand the attack surface.

1

Cloud

shared infrastructure and remote access

2

Endpoints

phones, laptops, tablets, sensors

3

Network

communication paths and exposed services

Challenges of Digital Technology Adoption

Many failures are social and organizational, not purely technical.

1

Resistance to change

people may fear complexity, loss of control, or extra work

2

Complicated tech

systems may be difficult to learn or poorly integrated

3

Unclear process

automation without redesign can preserve old problems

4

Risk and governance

security, privacy, law, and ethics need ownership

Adoption Needs Change Management

Technology adoption is a learning journey, not only installation.

1. **Explain why**
2. **Train clearly**
3. **Pilot safely**
4. **Measure use**
5. **Improve continuously**

Good adoption design answers: What changes? Who is affected? What support is needed? How will success be measured?

SECTION 4

Platforms and Networks

A short preview of the technical foundation behind digital services.



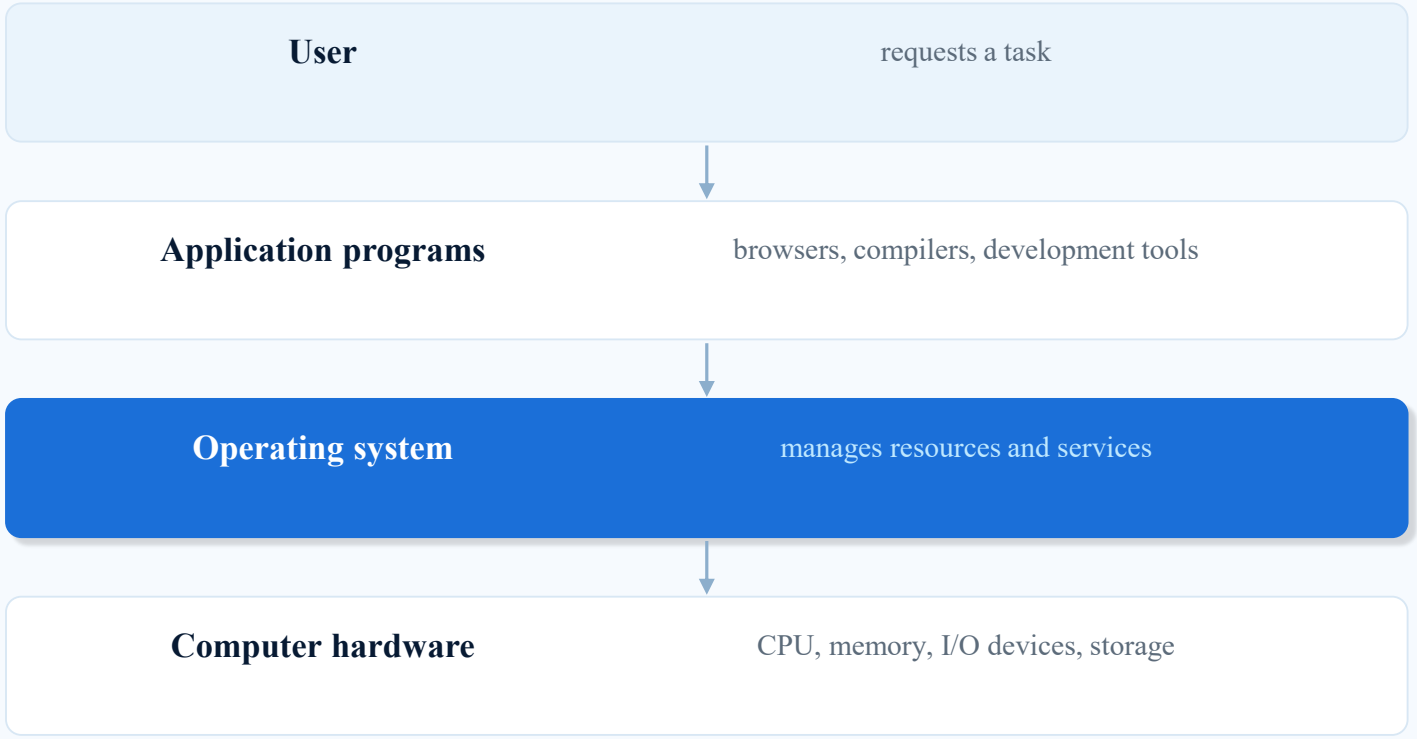
Computer Platform Overview

A platform combines hardware, operating system, applications, and supporting services.

Hardware	CPU • memory • storage • I/O
Operating system	resource manager • interface • services
Applications	user tasks • workflows • business logic
Network / Cloud	connectivity • scale • shared resources

Hardware + Software Layer Model

Students should recognize the basic stack before studying operating systems.



Operating System Roles

The operating system is the coordinator between applications and hardware.

1

Resource manager

allocates CPU, memory, storage, and devices

2

User interface

provides command line, desktop, mobile UI, or APIs

3

Security boundary

controls users, permissions, processes, and files

Virtual Machines and Cloud Computing

Virtualization separates software environments from physical machines.

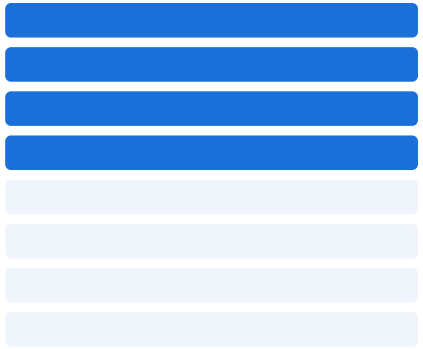
- Virtual machine: a software-defined computer running on shared hardware
- Cloud computing: on-demand computing resources delivered over networks
- Why it matters: flexibility, scalability, cost models, shared management



Cloud Service Models

Service models differ by who manages each layer.

IaaS



You manage OS, runtime, apps, data

PaaS



You manage apps and data

SaaS

Provider manages the full service

Computer Network Overview

A network links devices so they can exchange data and share services.

1

Devices

clients, servers, routers, sensors

2

Links

wired, wireless, fiber, cellular

3

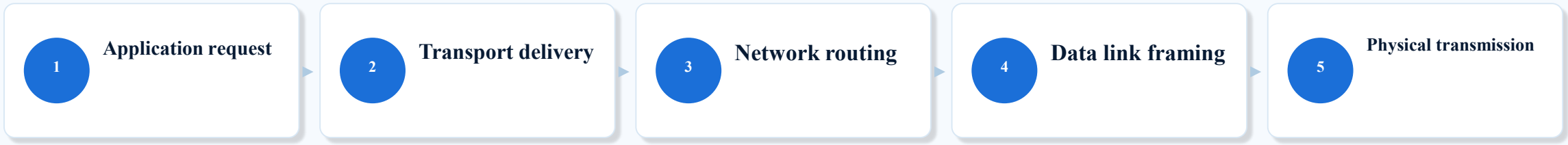
Protocols

rules for communication

Key questions: What is connected? How do they communicate? How are they protected?

Communication Protocols

Protocols are agreed rules that allow heterogeneous systems to communicate.



- Without protocols, devices may connect physically but fail to understand each other.
- Protocols structure data, addresses, reliability, timing, and error handling.
- Examples in later topics include HTTP, TCP, IP, Ethernet, Wi-Fi, DNS.

Network Security and CIA

Security goals help us reason about what protection means.

C

Confidentiality

protect data from unauthorized access

I

Integrity

ensure data accuracy and reliability

A

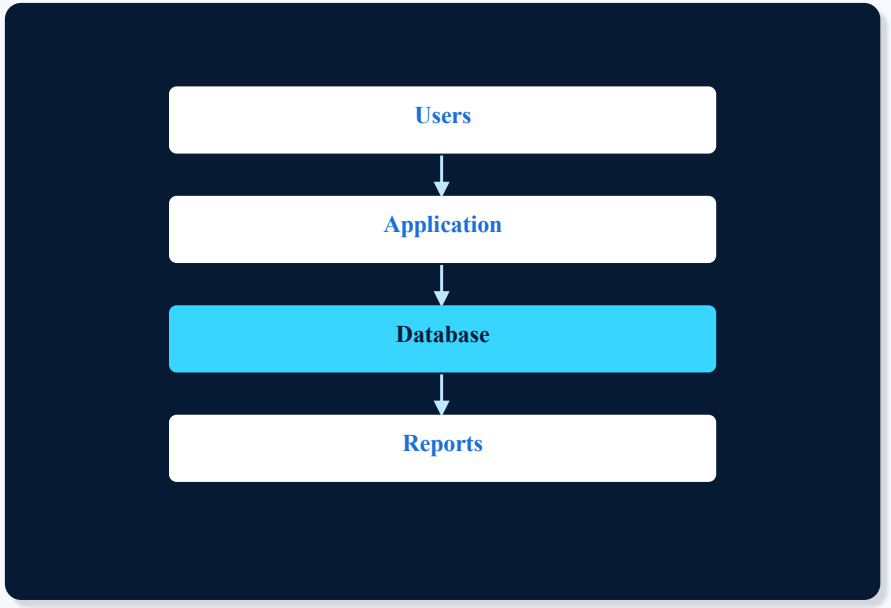
Availability

ensure data is accessible when needed

Database Systems

Databases organize data so services can store, retrieve, and protect information.

- Table / document / graph structures organize records
- Queries retrieve and update data
- Transactions keep important operations reliable
- Design affects performance, integrity, privacy, and reporting



Software Development and UX/UI

Digital services are designed, built, tested, deployed, and maintained.

1. **Requirements**
2. **Design**
3. **Development**
4. **Testing**
5. **Deployment**
6. **Maintenance**

UX/UI focus

Good software solves user problems with clear flows, feedback, accessibility, and trust.

Engineering focus

Good software is reliable, maintainable, secure, and measurable.

Emerging Digital Technologies

AI, blockchain, IoT, and automation extend what services can do.

1. **ChatGPT / GenAI**
   generate, summarize, code, explain

2. **Blockchain**
   shared ledgers and audit trails

3. **IoT**
   connected sensors and devices

4. **Automation**
   repeatable decisions and workflows

Law, Ethics, Academic Integrity, and AI Use

Digital work has responsibilities, not only capabilities.

Privacy	collect only what is needed and protect it
Security	prevent harm and unauthorized access
Fairness	watch for bias and exclusion
Integrity	cite sources; plagiarism means zero
AI use	use transparently according to course policy

Project and Presentation Preview

Students will analyze a problem and propose digital technology or services.

1. **Problem**
2. **Objectives**
3. **Conceptual ideas**
4. **Digital services**
5. **Conclusion**

- Mid-term: problem, objectives, conceptual ideas, benefits
- Final: include digital technologies/services and analysis
- Evaluate usability, accessibility, security, privacy, economy, society, environment

Activity 2: Analyze a Digital Service

Apply the building blocks model to one familiar service.

Choose one service: food delivery, online banking, learning platform, ride hailing, streaming, or e-commerce.

1

Users

Who uses it and who is affected?

2

Data

What data does it collect or create?

3

Software

What functions must it perform?

4

Network

What connections are required?

5

Risk

What could go wrong?

Week 1 Recap

A concise mental model for the rest of the course.

Technology	practical use of knowledge
Digital technology	electronic systems that process data
Digital services	value delivered through connected platforms
Adoption	people, process, risk, and governance
Foundation	platforms, OS, cloud, networks, security, data, software

Quick Check

Five questions to confirm understanding.

1. Give one example of technology that is not digital.
2. What makes a service “digital”?
3. Name two building blocks of a digital service.
4. Why is cloud computing important to digital services?
5. What are the three goals in CIA security?

NEXT

Next Week: Computer Platform

Bring one example of a digital service you use every day. Be ready to explain its hardware, software, data, network, and user value.

Thank you